

WEEK 3 – Day1

AFL DATASET ANALYSIS AND PREDICTIVE MODELING

1. Introduction

Sports analytics has become an important application of data science because modern sports generate large amounts of structured and unstructured data. Performance information collected from matches and players can be used to understand historical trends, evaluate teams, analyze individual performance, and develop predictive systems. Australian Football League (AFL) matches provide a suitable environment for applying these techniques because each match produces detailed information about teams, players, scoring, performance, opponents, venues, and match outcomes.

This project focuses on the analysis of historical AFL team and player data and the development of a machine learning system for predicting match outcomes. The project was designed as a complete data science workflow rather than simply training a machine learning model. The workflow begins with understanding the available datasets and defining an appropriate prediction target. It then continues with exploratory data analysis, feature engineering, data leakage prevention, predictive modeling, model evaluation, feature importance analysis, and model saving.

The main prediction problem addressed in this project is the classification of an AFL match into one of three possible outcomes: a home-team win, an away-team win, or a draw. In addition to match-level information, player performance statistics were incorporated into the feature engineering process so that the final predictive dataset could represent both team-level and player-level performance.

The overall purpose of the project is to demonstrate how historical sports data can be transformed into meaningful machine learning features and then used to build a practical predictive analytics system.

2. Project Scenario

The project is based on a realistic sports analytics scenario in which an AFL sports organization wants to use historical match and player information to improve its understanding of team performance and future match outcomes.

The organization has access to a large historical database containing match records for AFL teams as well as detailed statistics for individual players. However, raw statistics alone are difficult to use directly for prediction because they describe individual events and performances rather than providing a structured representation of the strength and recent form of the competing teams.

The proposed solution is therefore to develop a machine learning-based AFL analytics system. The system takes historical match and player information, processes the data, creates historical and recent performance indicators, compares the two competing teams, and uses these engineered features to predict the result of a future match.

The system is intended to answer questions such as how strongly a team's historical win rate affects its future performance, whether recent form is more informative than long-term performance, whether playing at home provides an advantage, how an opponent's historical strength influences the prediction, and whether aggregated player statistics can improve the representation of team performance.

The project therefore combines descriptive analytics and predictive analytics. Exploratory analysis is used to understand what has happened historically, while machine learning is used to learn patterns from the historical data and apply those patterns to future match predictions.

3. Problem Statement

The raw AFL datasets contain a large number of records and variables describing teams, matches, and players. However, these raw variables cannot simply be given to a machine learning model without understanding their meaning and relationship to the prediction target.

One of the major challenges is that match prediction must be based on information that would actually be available before the match takes place. For example, the final score of a match is highly informative about whether the team won, but that information would obviously not be available before the match. Using such information as a prediction feature would create data leakage and produce unrealistic results.

Another challenge is that team performance changes over time. A team's overall historical win rate may describe its long-term strength, but it may not accurately represent its current form. A team that performed well several seasons ago may currently be weaker, while a previously weaker team may currently be performing very well. For this reason, both historical and recent-performance features were developed.

The project therefore addresses the problem of transforming raw historical AFL data into a leakage-free machine learning dataset containing meaningful features that represent team strength, recent form, opponent strength, relative performance, home advantage, and player performance.

4. Dataset Description

Two primary datasets were used in the project. The first dataset contained team-level match information, while the second contained detailed round-by-round player statistics.

The team match dataset was stored in the file `team_matches_home_away_raw - team_matches_home_away_raw.csv.csv`. After loading the dataset, it contained 15,808 records

and 19 columns. The dataset provided information about the team involved in each record, its opponent, the match date, round, home or away status, scores, result, margin, venue, crowd, and goal-related statistics.

The team dataset was particularly important because it provided the basic structure required for match-level analysis. It allowed the project to determine which team was playing, who the opponent was, whether the team was playing at home, how many points were scored, how many points were conceded, and what the final result of the match was.

The second dataset was stored in `afl_players_round_by_round_stats_raw - afl_players_round_by_round_stats_raw.csv.csv`. This dataset contained 274,089 player-level records and 36 columns. It contained detailed information about player performance, including kicks, marks, handballs, disposals, goals, behinds, hit-outs, tackles, rebound 50s, inside 50s, clearances, clangers, free kicks, Brownlow votes, contested possessions, uncontested possessions, contested marks, marks inside 50, one percenters, bounces, goal assists, percentage of game played, fantasy points, and other performance-related variables.

The player dataset provided a more detailed view of team performance because individual player contributions could be aggregated to represent the overall player strength available to a team before a match.

5. Tools and Technologies Used

Python was used as the main programming language for the complete project because it provides a strong ecosystem for data analysis, visualization, and machine learning.

Pandas was used extensively for loading CSV files, inspecting datasets, manipulating columns, grouping records, sorting historical observations, calculating statistics, and creating the final modeling dataset. NumPy was used for numerical operations and safe handling of mathematical calculations during feature engineering.

Matplotlib was used to create visualizations during the exploratory data analysis stage. These visualizations helped understand team performance, match outcomes, player performance, relationships between variables, and model performance.

Scikit-learn was used during the machine learning stage. It provided the implementations of LabelEncoder, StandardScaler, Logistic Regression, Decision Tree, Random Forest, and the evaluation metrics used to compare the models.

Joblib was used at the end of the project to save the trained machine learning models so that they could be loaded and reused later without repeating the complete training process.

6. Task 1 Data Understanding and Preparation

The first task focused on understanding the structure and content of the AFL datasets before performing any machine learning. This stage was important because predictive modeling should only begin after the data has been properly understood.

The team match dataset and player statistics dataset were loaded using Pandas. After loading the files, their shapes were examined to determine the number of observations and available variables. The team dataset contained 15,808 rows and 19 columns, while the player dataset contained 274,089 rows and 36 columns.

The column names were then inspected to understand the meaning of the available variables. The team dataset contained fields such as team name, opponent, round, match date, home or away status, team score, opponent score, result, margin, venue, crowd, and goal statistics. The player dataset contained detailed player performance variables such as disposals, goals, tackles, marks, fantasy points, and other AFL statistics.

The match date was converted into a datetime format so that chronological operations could be performed correctly. This was especially important because the project later required historical and rolling statistics. Without a proper datetime representation, it would not be possible to reliably sort matches according to when they occurred.

The initial data understanding stage also helped determine which variables could be used for target creation, exploratory analysis, feature engineering, and prediction. At this stage, the project established the basic relationship between the team-level data and player-level data.

The result of Task 1 was a clear understanding of the datasets and their available information, providing the foundation for the next stage of the project.

7. Task 2 Target Definition and Match-Level Labels

The second task focused on defining the target that the machine learning models would eventually predict. Since the main objective of the project was match prediction, the primary target was defined as `match_result`.

The match result was divided into three possible classes. A match was classified as `home_win` when the home team scored more points than its opponent. A match was classified as `away_win` when the away team scored more points than its opponent. If both teams finished with the same score, the match was classified as a draw.

This created a three-class classification problem. Instead of predicting a numerical score directly, the machine learning models would learn to determine which of the three possible outcomes was most likely.

The player dataset was also examined for individual match performance. Two concepts, `top_disposals` and `top_goals`, were considered for identifying the player with the highest number of disposals or goals within a match. This analysis helped connect individual player performance with the overall match context.

One practical issue encountered during target and match-level analysis was that the player dataset did not provide a simple unique `match_id` that could be used to identify every match. To overcome this problem, match records were associated using available fields such as match date, round, team, and opponent. This provided a practical way to group player records into match-level observations.

The target-definition stage was therefore important because it transformed the raw match information into a clearly defined supervised learning problem.

8. Task 3 — Exploratory Data Analysis

The third task focused on exploratory data analysis. The purpose of EDA was to understand historical AFL performance patterns before building predictive models.

The first area examined was team performance. Historical win rates were calculated to compare the performance of different teams. This allowed the analysis to identify teams that had historically achieved stronger results and teams that had experienced weaker performance.

The distribution of match outcomes was also analyzed. Home wins, away wins, and draws were examined to understand how frequently each type of outcome occurred in the historical dataset. This analysis was particularly important because the target variable used later in machine learning was based on these three classes.

Home and away performance was then investigated to understand whether teams generally performed differently depending on where the match was played. The home or away status was considered an important factor because playing at a familiar venue and in front of a home crowd may influence team performance.

Season-level trends were also analyzed. Average team scores, average opponent scores, and match margins were examined across seasons. This provided a historical perspective on how scoring and competitiveness changed over time.

Player performance was analyzed using statistics such as disposals, goals, tackles, marks, and fantasy points. Distributions of these variables helped identify typical player performance levels and extreme performances. Historical leaders were also examined based on total disposals and total goals.

Player consistency was analyzed using the variation in disposals. Standard deviation was used as an indicator of how consistently a player performed across matches. A lower variation represented more stable performance, while a higher variation represented greater fluctuation.

Relationships between player performance variables were also examined. Scatter plots and correlation analysis were used to investigate relationships between disposals, goals, tackles, and fantasy points. This provided additional information about which variables might contain useful predictive information.

The EDA stage provided important insights that guided the feature-engineering stage. Historical team strength, recent performance, scoring, defensive performance, home advantage, and player performance all appeared to be potentially useful for predicting future match outcomes.

9. Task 4 — Feature Engineering

Feature engineering was one of the most important stages of the entire project. The objective was to transform raw AFL data into meaningful predictive variables that represented the information realistically available before a match.

The first engineered feature was `is_home`. This feature represents whether the team was playing at its home ground. A value of one represented a home match, while zero represented an away match. This allowed the model to learn whether home-ground advantage contributed to match outcomes.

The number of previous matches played by each team was calculated using the `previous_matches` feature. This represented how much historical match experience was available for a team at the time of a particular match.

A cumulative historical win rate was then calculated using `previous_win_rate`. This feature represented how successful a team had been in its previous matches. Importantly, the current match was excluded from this calculation so that the feature would not contain information from the outcome being predicted.

Recent form was represented using `recent_win_rate_5`, which captured the team's win rate across its previous five matches. This feature was important because recent performance can provide a better indication of current team strength than a long-term historical average alone.

Historical scoring performance was represented using `previous_avg_score`, while defensive performance was represented using `previous_avg_conceded`. These features described the average number of points a team had scored and conceded before the current match.

Recent scoring performance was captured using `recent_avg_score_5` and `recent_avg_conceded_5`. These rolling five-match statistics provided a more current representation of offensive and defensive form.

Match margin was also incorporated into the feature set. The `previous_avg_margin` feature represented historical average score difference, while `recent_avg_margin_5` represented recent average margin. These features provided an additional indication of how strongly a team had been outperforming or underperforming its opponents.

10. Opponent Feature Engineering

A major improvement in the feature-engineering process was the inclusion of opponent information.

A team's probability of winning cannot be evaluated only from its own historical performance. The strength of the opposing team also has a major effect on the outcome. For example, a team with a strong historical record may still have difficulty winning when facing another highly successful team.

For this reason, opponent-specific historical and recent features were created. These included opponent historical win rate, opponent recent win rate, opponent historical scoring performance, opponent historical defensive performance, opponent recent scoring performance, opponent recent defensive performance, and opponent historical and recent margins.

These features allowed the machine learning models to understand the quality of the opponent rather than treating every match as if the opposition had the same strength.

11. Relative Strength Feature Engineering

After creating team and opponent features separately, relative strength features were created to directly compare the two teams.

The `win_rate_difference` feature represented the difference between the team's historical win rate and the opponent's historical win rate. The `recent_form_difference` feature compared the recent five-match performance of the two teams.

Similarly, `scoring_strength_difference` compared the scoring strength of the competing teams, while `defensive_strength_difference` compared their defensive performance. The `recent_scoring_difference` feature compared their recent scoring ability.

These relative features were particularly useful because match prediction is fundamentally a comparison problem. The model does not only need to know whether a team is strong; it needs to know whether it is stronger or weaker than the opponent it is facing.

12. Player-Level Feature Engineering

Player statistics were incorporated into the team-level prediction dataset by aggregating individual player performance.

Several important player statistics were selected, including disposals, goals, tackles, marks, and fantasy points. These statistics were aggregated so that the overall player performance associated with a team could be represented at the match level.

Historical averages were calculated for player disposals, goals, tackles, and fantasy points. These features represented the team's historical player-performance level.

Recent five-match averages were also calculated for disposals, goals, tackles, and fantasy points. These features represented the team's recent player performance.

The use of player-based features allowed the model to capture an additional dimension of team strength. Instead of considering only final scores and historical results, the model could also learn from the underlying performance of the players contributing to the team.

13. Data Leakage Prevention

Data leakage prevention was a critical part of feature engineering.

A machine learning model should only use information that would realistically be available at prediction time. Current-match information such as the final team score, opponent score, final margin, current match result, current goals, and current behinds was therefore excluded from the predictive feature set.

Using such variables would make the prediction problem artificially easy because these variables are directly related to the result being predicted and are only known after the match has taken place.

Instead, the project used previous-match statistics, historical averages, rolling five-match averages, opponent historical performance, and player performance from earlier matches.

This approach made the feature set more realistic and reduced the risk of the model learning information that would not actually be available when predicting a future AFL match.

14. Final Engineered Dataset

After completing feature engineering, a final set of 32 predictive features was selected.

These features represented home advantage, historical team strength, recent form, scoring ability, defensive performance, historical and recent opponent strength, relative team strength, and player performance.

The final feature set was stored in the engineered dataset `AFL_engineered_features.csv`.

Saving the engineered dataset was particularly useful later in the project because restarting the notebook kernel removed previously created Python variables. Instead of repeating all feature-engineering operations, the saved engineered dataset could be loaded again and used to reconstruct the modeling environment.

15. Task 5 — Predictive Modeling

The fifth and final major task focused on developing machine learning models for match-result prediction.

The target variable `match_result` contained three classes: `away_win`, `draw`, and `home_win`. Since machine learning algorithms require numerical target values, `LabelEncoder` was used to convert these classes into numerical labels.

The engineered feature dataset was then sorted chronologically according to match date. This was an important decision because the objective was to simulate a realistic future prediction scenario.

Instead of randomly selecting training and testing records, an 80/20 chronological split was used. The earliest 80 percent of the matches were used for model training, while the latest 20 percent were reserved for testing.

This approach prevents future information from being used to train predictions for earlier matches and provides a more realistic evaluation of how the models could perform on future AFL matches.

16. Logistic Regression Model

The first machine learning algorithm used was Logistic Regression.

Logistic Regression was selected as a baseline classification model because it provides a relatively simple approach for classification and allows the performance of more complex models to be compared against a basic predictive method.

A `StandardScaler` was included before the Logistic Regression classifier. Feature scaling was important because the engineered variables could have different numerical ranges. Standardization places the variables on comparable scales and allows the optimization process to work more effectively.

The Logistic Regression model was trained using the 32 engineered features and the encoded match-result target.

17. Decision Tree Model

The second algorithm used was a Decision Tree Classifier.

Decision Trees were selected because they can capture nonlinear relationships between features and match outcomes. Unlike Logistic Regression, a Decision Tree can divide the feature space into different decision regions based on conditions involving the engineered variables.

The tree was configured with a maximum depth of eight and a minimum split requirement to reduce excessive complexity. These restrictions were applied to reduce the risk of overfitting.

The Decision Tree was trained using the same chronological training data and evaluated on the same testing data as the other models.

18. Random Forest Model

The third algorithm used was Random Forest.

Random Forest is an ensemble learning algorithm that combines multiple decision trees to produce a more robust prediction. Instead of depending on a single tree, the algorithm creates many trees and combines their predictions.

The Random Forest model was configured with 200 estimators, a maximum depth of 10, a minimum split requirement of 10, and a fixed random state for reproducibility. Parallel processing was enabled to improve training efficiency.

Random Forest was particularly useful for this project because the relationship between team performance, recent form, opponent strength, and player statistics is likely to be nonlinear and may involve interactions between multiple features.

19. Model Evaluation

After training the three models, their predictions were generated on the test dataset.

Several evaluation metrics were used to compare their performance. Accuracy was used to determine the overall percentage of correctly classified matches. Precision was used to measure how reliable the predictions were for each class. Recall was used to determine how effectively the models identified the actual examples belonging to each class.

The F1 Score was also calculated because it combines precision and recall into a single measure. A weighted F1 Score was used so that the evaluation considered the distribution of the different target classes.

Using multiple metrics was important because accuracy alone does not provide a complete picture of classification performance. A model could achieve reasonable overall accuracy while performing poorly on one particular class. Precision, recall, and F1 Score therefore provided a more detailed evaluation.

20. Classification Reports

Classification reports were generated for Logistic Regression, Decision Tree, and Random Forest.

Each report provided class-level precision, recall, F1 Score, and support for the three match-result categories.

This allowed the performance of each model to be examined separately for home wins, away wins, and draws.

The classification reports were useful for identifying whether the models were equally capable of predicting all three classes or whether certain outcomes were more difficult to identify.

21. Confusion Matrix Analysis

Confusion matrices were generated for each of the three models.

A confusion matrix provides a visual representation of correct and incorrect predictions for each target class. The rows represent the actual classes while the columns represent the predicted classes.

The confusion matrices allowed the project to identify how frequently home wins were confused with away wins, how frequently draws were incorrectly classified, and which match-result category was the most difficult for the models.

This analysis complemented the numerical evaluation metrics and provided a more detailed understanding of model behavior.

22. Model Comparison

The performance of Logistic Regression, Decision Tree, and Random Forest was compared using accuracy, precision, recall, and F1 Score.

A model comparison dataframe was created to organize the evaluation results. A visualization was also generated to compare the models across the different metrics.

The F1 Score was selected as the main criterion for identifying the best-performing model because it provides a balance between precision and recall.

The model with the highest F1 Score was therefore selected as the best model for the developed AFL prediction system.

The exact numerical values of the model comparison were generated directly from the notebook after executing the evaluation cells.

23. Random Forest Feature Importance

Feature importance analysis was performed using the Random Forest model.

The purpose of this analysis was to understand which engineered variables contributed most strongly to the model's predictions.

The feature-importance results were sorted from highest to lowest importance. The top features were then visualized using a bar chart.

This analysis provided an additional level of interpretation because the project was not only interested in predicting the result but also in understanding which aspects of team and player performance were important for prediction.

The feature-importance analysis could therefore help sports analysts identify which historical, recent, opponent, and player-related factors deserve greater attention.

24. Saving the Models and Results

After model training and evaluation, the trained machine learning models were saved using Joblib.

The Logistic Regression model was saved as `AFL_logistic_model.joblib`. The Decision Tree model was saved as `AFL_decision_tree_model.joblib`, while the Random Forest model was saved as `AFL_random_forest_model.joblib`.

The model comparison results were saved as `AFL_model_comparison.csv`, allowing the evaluation results to be accessed later without rerunning the complete evaluation process.

The feature-importance results were saved as `AFL_feature_importance.csv`, allowing the most important predictive features to be reviewed independently.

The engineered dataset was also preserved as `AFL_engineered_features.csv`.

This approach makes the project reproducible and allows the trained models to be integrated into a future prediction application.

25. Problems Encountered During Implementation

Several practical problems were encountered while implementing the project.

One of the main technical issues occurred when Scikit-learn was not installed in the active Python environment. When the machine learning import cell was executed, the notebook produced a `ModuleNotFoundError` indicating that the `sklearn` module was missing. The problem was resolved by installing the Scikit-learn package and restarting the notebook kernel.

Restarting the kernel introduced another issue because all previously created Python variables were removed from memory. Variables such as `team_features`, `X`, `y`, and `final_feature_columns` were therefore no longer available.

Instead of repeating the entire feature-engineering process, the previously saved `AFL_engineered_features.csv` file was loaded. The required feature columns were reconstructed from the saved dataset, allowing the machine learning stage to continue.

Another issue occurred during the historical win-rate calculation. When a team had no previous matches, dividing previous wins by previous matches could result in a division-by-zero problem. The calculation was therefore modified so that zero previous-match values were safely handled before calculating the win rate.

These implementation challenges provided practical experience in debugging, environment management, data persistence, and safe feature engineering.

26. Complete Project Workflow

The complete project followed a structured data science pipeline.

The process began by loading the raw AFL team and player datasets. The datasets were then inspected to understand their dimensions, columns, and available information.

After understanding the data, the match-result target was defined using home wins, away wins, and draws. Exploratory data analysis was then performed to understand team performance, match outcomes, home advantage, season trends, player performance, player consistency, and relationships between statistical variables.

The next stage focused on feature engineering. Historical team performance, recent form, scoring performance, defensive performance, opponent strength, relative strength, and player performance were transformed into 32 predictive features.

Data leakage prevention was applied by ensuring that current-match outcome information was not included in the predictive features.

The engineered dataset was then sorted chronologically and divided into training and testing datasets using an 80/20 chronological split.

Three machine learning models were trained: Logistic Regression, Decision Tree, and Random Forest. Their performance was evaluated using accuracy, precision, recall, F1 Score, classification reports, and confusion matrices.

Finally, the models were compared, the best-performing model was identified using F1 Score, Random Forest feature importance was analyzed, and the trained models and results were saved for future use.

27. Key Findings

The exploratory analysis demonstrated that AFL team performance can be represented through multiple dimensions rather than a single statistic.

Historical win rate provides information about long-term team strength, while recent win rate provides information about current form. Historical and recent scoring statistics provide information about offensive strength, while conceded points and margin provide information about defensive and overall competitive performance.

Opponent information is also important because the difficulty of a match depends on the strength of the opposing team. Relative features therefore provide a more meaningful representation of the prediction problem than team statistics alone.

Player statistics provide another important dimension. Aggregated disposals, goals, tackles, and fantasy points can represent the performance level of the players contributing to the team.

The feature-engineering process therefore created a richer representation of each match than the original raw dataset.

29. Future Improvements

The project could be extended by integrating player availability information. Injury status, suspension information, selected players, and last-minute withdrawals could provide valuable additional information before a match. Weather information could also be incorporated. Variables such as temperature, rainfall, wind speed, and weather conditions could potentially improve predictions. Travel distance and rest days could also be considered because teams may perform differently depending on how far they have travelled and how much recovery time they have between matches. The machine learning component could also be improved through hyperparameter optimization. Grid Search or Randomized Search could be used to identify better combinations of model parameters. Additional algorithms such as Gradient Boosting, XGBoost, LightGBM, or neural networks could also be evaluated. A future version could further develop the project into a complete web-based sports analytics application where users could select two teams and receive a predicted match outcome along with the most influential factors behind the prediction.

30. Conclusion

This project successfully developed an end-to-end AFL sports analytics and machine learning workflow using historical team and player statistics. The project started with two large raw datasets containing team-level match information and detailed player-level performance

statistics. The data was examined and prepared so that it could be transformed into a suitable supervised learning problem. The match-result target was defined using three possible outcomes: home win, away win, and draw. Exploratory data analysis was then performed to understand historical team performance, match outcomes, home advantage, seasonal scoring patterns, player performance, player consistency, and relationships between important statistical variables. Feature engineering transformed the raw information into a structured set of 32 predictive features. These features represented historical team strength, recent form, scoring performance, defensive performance, opponent strength, relative team strength, home advantage, and aggregated player performance. A major focus of the project was preventing data leakage. Current-match information that would not be available before the match was excluded from the predictive feature set. Instead, historical and rolling statistics were used to simulate a realistic future prediction environment.

Three machine learning algorithms were then trained: Logistic Regression, Decision Tree, and Random Forest. A chronological 80/20 train-test split was used to ensure that the models learned from earlier matches and were evaluated on later matches.

The models were evaluated using accuracy, precision, recall, and F1 Score. Classification reports and confusion matrices were also generated to provide a detailed understanding of model performance. Random Forest feature importance was analyzed to understand which engineered variables contributed most strongly to predictions.

The project also demonstrated practical aspects of machine learning development, including package installation, kernel management, recovery of saved engineered data, handling division-by-zero situations, model persistence, and result saving.

Overall, the project demonstrates that AFL historical data can be transformed into a meaningful machine learning dataset capable of supporting match-result prediction. More importantly, it demonstrates the complete process required to move from raw sports data to an interpretable predictive system.

The developed system provides a strong foundation for future improvements. By incorporating player availability, injuries, weather conditions, travel information, rest periods, advanced algorithms, and hyperparameter optimization, the predictive performance and practical usefulness of the system could be further improved.

The final outcome of the project is therefore not simply a trained machine learning model, but a complete sports analytics pipeline covering data understanding, target definition, exploratory analysis, feature engineering, leakage prevention, predictive modeling, evaluation, interpretation, and model persistence.